SAL COLLEGE AND ENGINEERING

Analysis and Design of Algorithms–(BE04000241)

SEM – 4 B Branch – Computer Engineering

Greedy Algorithm

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Introduction to Greedy Algorithm

A Greedy Algorithm is an algorithmic strategy that builds a solution step by step by choosing the best option available at the current moment.

- It selects the locally optimal choice at each step.
- The goal is to reach a globally optimal solution.
- Used mainly for optimization problems.

Example idea:

- Always take the best immediate benefit.

Key Idea of Greedy Algorithms

The greedy approach works by repeatedly making the best choice at the moment without re-considering previous decisions.

Main concept:

Local optimum → hope to reach global optimum.

Example analogy:

Selecting highest value coin first in coin change problem.

Characteristics of Greedy Algorithms

Greedy algorithms usually have these properties:

1. Greedy Choice Property
   - The best local choice leads to the optimal solution.

1. Optimal Substructure
   - Optimal solution can be constructed from optimal solutions of subproblems.

General Structure of Greedy Algorithm

Greedy algorithms follow a simple structure:

1. Initialize solution set
2. Choose the best candidate
3. Add candidate to solution
4. Repeat until solution is complete

Instead of checking all possible combinations, the algorithm focuses only on the best choice at each step.

Applications of Greedy Algorithms

Greedy algorithms are used in many well-known problems:

- Activity Selection Problem
- Fractional Knapsack
- Minimum Spanning Tree
  - Kruskal’s Algorithm
  - Prim’s Algorithm
- Huffman Coding
- Dijkstra’s Shortest Path

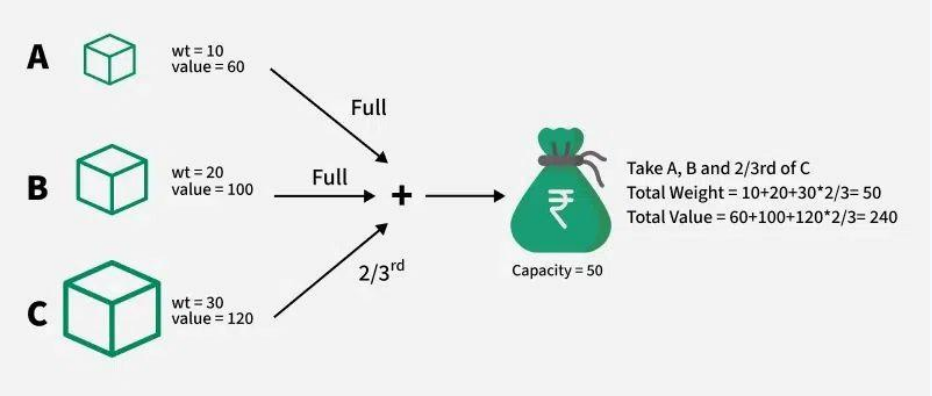
They are commonly used in optimization and scheduling problems.

Fractional Knapsack Problem

Problem statement:

Given:

- Items with value and weight
- A knapsack with limited capacity



Goal:

- Maximize the total value inside the knapsack.

In Fractional Knapsack, items can be broken into fractions to maximize value.

Greedy Approach for Fractional Knapsack

Steps:

1. Calculate value/weight ratio for each item.
2. Sort items based on highest ratio.
3. Pick items with highest ratio first.
4. If the next item doesn't fit completely, take fraction of it.

This ensures maximum profit in the knapsack.

Example of Fractional Knapsack

Item	Weight	Value	Ratio
1	10	60	6
2	20	100	5
3	30	120	4

Steps:

- Select item with highest ratio first.
- Continue until capacity is full.

Result → Maximum possible value.

Advantages of Greedy Algorithm

Advantages:

- Simple to understand
- Easy to implement
- Efficient for many optimization problems
- Often faster than dynamic programming

Disadvantages of Greedy Algorithm

Disadvantages:

- Does not guarantee optimal solution for all problems
- Works only when greedy property exists
- Sometimes dynamic programming is required instead

Real-World Applications

Greedy algorithms are used in:

- Network routing
- Data compression (Huffman coding)
- Scheduling tasks
- Resource allocation

These algorithms are popular because they produce good results quickly.

Conclusion

- Greedy algorithms choose the best immediate solution at every step.
- They are widely used for optimization problems.
- Example: Fractional Knapsack, Activity Selection.
- Although simple and efficient, they do not always guarantee the optimal solution.

Thank you